

Volatility Exercise

1. If t is a time period expressed in years, σ is the annual volatility, and F is the forward or futures price, then for simple volatility calculations (rather than exponential calculations) a price change of n standard deviations over the time period t can be approximated as

$$n \times F \times \sigma \times \sqrt{t}$$

For a daily standard deviation, traders customarily assume 256 trading days in a year. t is therefore $1/256$, and $\sqrt{t} = 1/16$

For a weekly standard deviation, assuming 52 trading weeks in a year, t is $1/52$, and $\sqrt{t} \approx 1/7.2$.

For a monthly standard deviation, assuming 12 months in a year, t is $1/12$, and $\sqrt{t} \approx 1/3.5$.

Using simple volatility calculations, and assuming that the forward price is equal to the current price, what is an approximate daily, weekly, and monthly one standard deviation price change for each of the contracts on the following page?

Volatility Exercise

1 a.) stock price = 78.00

volatility: 20% 24% 30% 38%

daily:

weekly:

monthly:

1 b.) index price = 4468.00

volatility: 7% 12% 20% 30%

daily:

weekly:

monthly:

Volatility Exercise

2. Using simple volatility calculations, for each underlying contract and volatility below, what is an approximate one and two standard deviation price range over the given time period? Assume that a year is made up of 12 months, 52 weeks, or 365 days.

	<u>underlying price</u>	<u>volatility</u>	<u>time period</u>	<u>down one st. dev.</u>	<u>up one st. dev.</u>	<u>down two st. devs.</u>	<u>up two st. devs.</u>
a.)	226.00	21%	14 weeks				
b.)	1869.00	14%	11 months				
c.)	103.82	9.5%	116 days				

Volatility Exercise

3. For a normal distribution, the following are the approximate probabilities associated with one, two, and three standard deviations:

a move greater than one standard deviation, in either direction $31\frac{3}{4}\%$

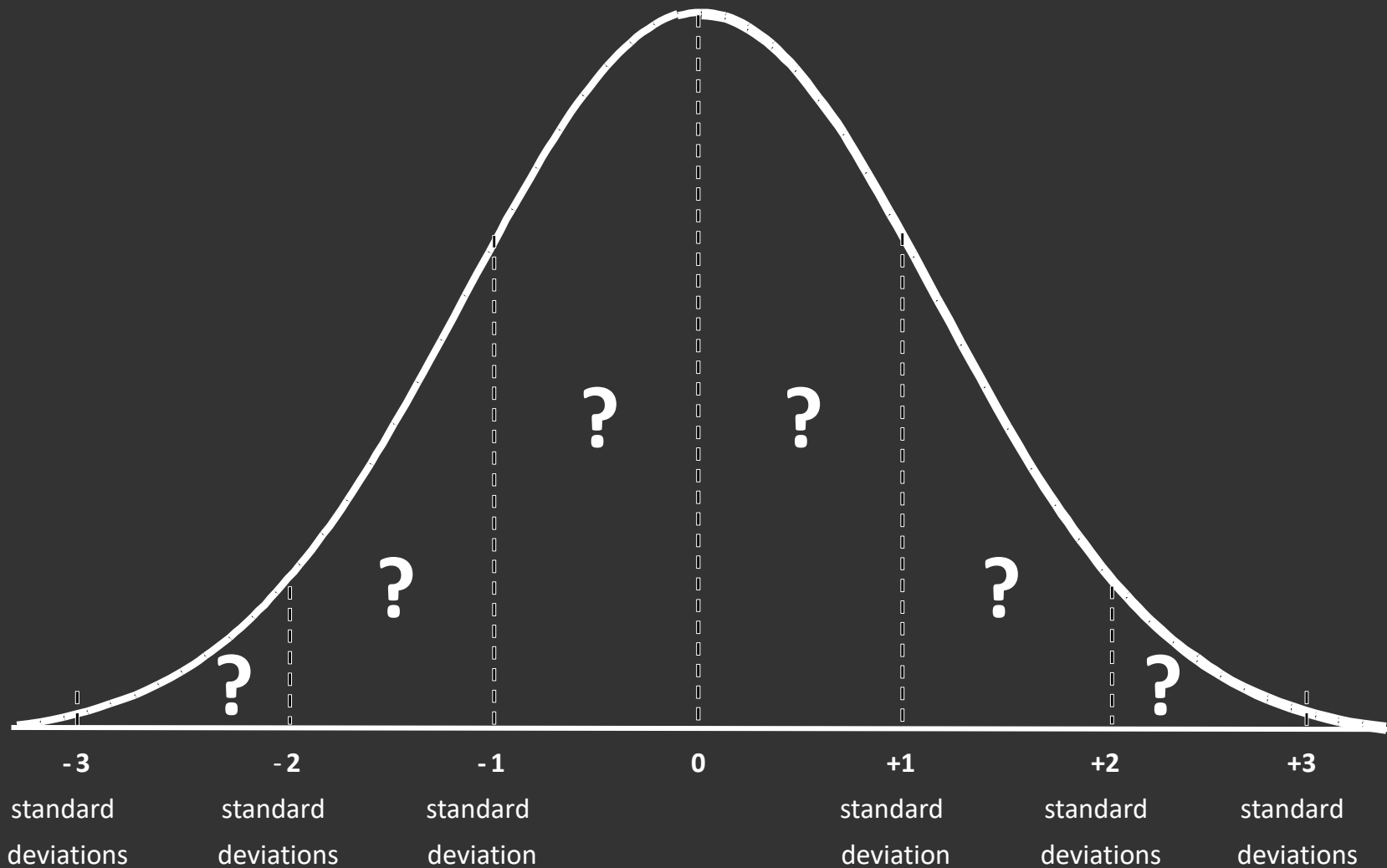
a move greater than two standard deviations, in either direction $4\frac{1}{2}\%$

a move greater than three standard deviations, in either direction $\frac{1}{4}\%$

Using the above values, fill in the approximate probabilities indicated by the question marks in the normal distribution on the following page (the percentage of the area under the normal distribution curve between each set of dotted lines).

Volatility Exercise

3.



Volatility Exercise

4. From your answers to question 3, what is the approximate probability of getting an occurrence over the following ranges:
- a.) An up move of less than two standard deviations
 - b.) An up or down move between one and two standard deviations
 - c.) A down move between two and three standard deviations
 - d.) An up move of less than one standard deviation or a down move of less than two standard deviations.
 - e.) A down move between one and two standard deviations and an up move between two and three standard deviations
 - f.) An up move greater than three standard deviations

Volatility Exercise

5. For each pair of options below, if there is a change in volatility which option's value will change the most in total points? Assume that the underlying contract is trading at a price of 50, and that all months are in the same calendar year.

a.) April 60 call
June 60 call

b.) April 60 put
June 45 put

c.) April 45 call
April 60 call

d.) April 45 put
June 50 put

e.) June 40 put
June 45 put

f.) April 45 call
June 50 call

g.) April 55 put
April 60 put

h.) April 55 call
June 50 call

i.) June 50 call
June 55 call

6. For each pair of options below, if there is a change in volatility which option's value will change the most in percent terms? Assume that the underlying contract is trading at a price of 75, and that all months are in the same calendar year.

a.) February 85 call
February 75 call

b.) May 80 put
May 85 put

c.) May 90 call
May 80 call

d.) May 75 put
May 70 put

e.) February 65 call
February 70 call

f.) February 85 put
February 75 put

1 a.) stock price = 78.00

volatility:	<u>20%</u>	<u>24%</u>	<u>30%</u>	<u>38%</u>
daily:	.98	1.17	1.46	1.85
weekly:	2.17	2.60	3.25	4.12
monthly:	4.46	5.35	6.69	8.47

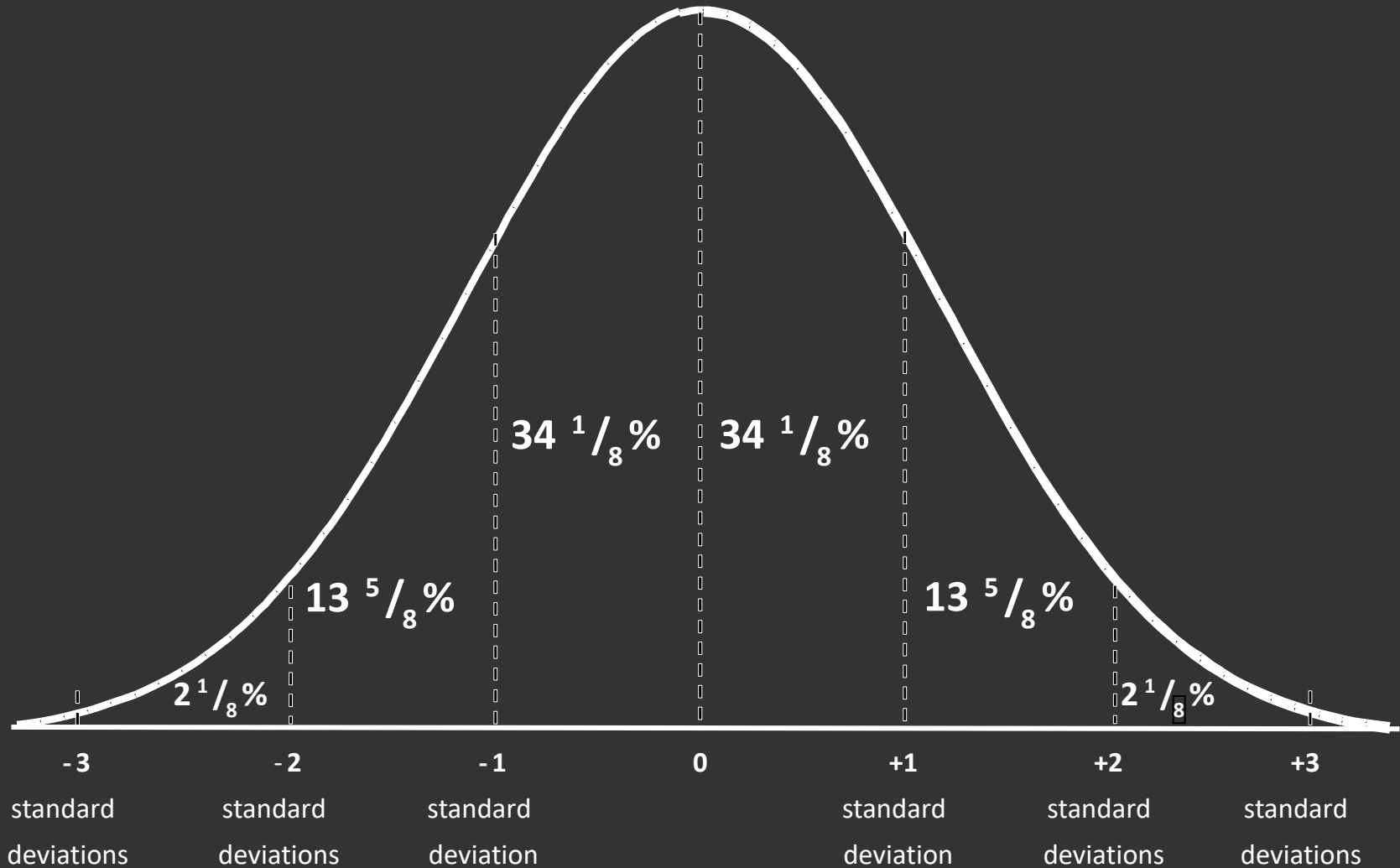
1 b.) index price = 4468.00

volatility:	<u>7%</u>	<u>12%</u>	<u>20%</u>	<u>30%</u>
daily:	19.55	33.51	55.85	83.78
weekly:	43.44	74.47	124.11	186.17
monthly:	89.36	153.19	255.31	382.97

2. Using simple volatility calculations, for each underlying contract and volatility below, what is an approximate one and two standard deviation price range over the given time period? Assume that a year is made up of 12 months, 52 weeks, or 365 days.

	<u>underlying price</u>	<u>volatility</u>	<u>time period</u>	<u>down one st. dev.</u>	<u>up one st. dev.</u>	<u>down two st. devs.</u>	<u>up two st. devs.</u>
a.)	226.00	21%	14 weeks	201.37	250.64	176.75	275.25
b.)	1869.00	14%	11 months	1618.48	2119.52	1367.96	2370.04
c.)	103.82	9.5%	116 days	98.25	109.36	92.70	114.94

3.



4. From your answers to question 3, what is the approximate probability of getting an occurrence over the following ranges:

a.) An up move of less than two standard deviations

$$34\frac{1}{8}\% + 13\frac{5}{8}\% = 47\frac{3}{4}\% \text{ (slightly less than 1 chance in 2)}$$

b.) An up or down move between one and two standard deviations

$$2 \times 13\frac{5}{8}\% = 27\frac{1}{4}\% \text{ (slightly more than 1 chance in 4)}$$

c.) A down move between two and three standard deviations

$$2\frac{1}{8}\% - \frac{1}{8}\% = 2\% \text{ (1 chance in 50)}$$

d.) An up move of less than one standard deviation or a down move of less than two standard deviations.

$$34\frac{1}{4}\% + 34\frac{1}{4}\% + 13\frac{1}{2}\% = 82\% \text{ (slightly more than 4 times out of 5)}$$

e.) A down move between one and two standard deviations and an up move between two and three standard deviations

$$13\frac{5}{8}\% + 2\frac{1}{8}\% - \frac{1}{8}\% = 16\% \text{ (about 1 chance in 6)}$$

f.) An up move greater than three standard deviations

$$\frac{1}{8}\% \text{ (about 1 chance in 800)}$$

5. For each pair of options below, if there is a change in volatility which option's value will change the most in total points? Assume that the underlying contract is trading at a price of 50, and that all months are in the same calendar year.

a.) April 60 call
June 60 call

b.) April 60 put
June 45 put

c.) **April 45 call**
April 60 call

d.) April 45 put
June 50 put

e.) June 40 put
June 45 put

f.) April 45 call
June 50 call

g.) **April 55 put**
April 60 put

h.) April 55 call
June 50 call

i.) **June 50 call**
June 55 call

6. For each pair of options below, if there is a change in volatility which option's value will change the most in percent terms? Assume that the underlying contract is trading at a price of 75, and that all months are in the same calendar year.

a.) **February 85 call**
February 75 call

b.) **May 80 put**
May 85 put

c.) **May 90 call**
May 80 call

d.) May 75 put
May 70 put

e.) February 65 call
February 70 call

f.) February 85 put
February 75 put

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